Tutorial Problem Set VIII

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Topics: related rates, linear approximations and the mean value theorem.

1 Basic problems

Problem 1. Boyle's law states that if P denotes the absolute pressure exerted by a given mass of an ideal gas and V denotes the total volume it occupies, then PV is constant provided that the temperature and amount of gas remains unchanged within a closed system.

(a) If the volume of the gas decreases linearly, does it necessarily follow that the pressure decreases linearly?

(b) Suppose that the pressure at a given point in time is 100 kPa and that the volume is 200 cm^3. Given that the volume decreases with a rate of 40 cm^3 per minute, compute the instantaneous rate of change of the pressure exerted by the gas.

Problem 2. The following problems are about approximations.

(a) Approximate $\log(1.1)$ using a linear approximation. Using $f(a+h) \approx f(a) + f'(a)h + \frac{f''(a)}{2}h^2$, find another approximation of $\log(1.1)$.

(b) More generally, the *nth Taylor polynomial* of a function $f : \mathbb{R} \to \mathbb{R}$ at a is defined as

$$T_n(x) = f(a) + f'(a)h + \frac{f''(a)}{2}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Show that the kth derivative of $T_n(x)$ at a is equal to $f^k(a)$ if $k \leq n$.

(c) Compute the Taylor polynomials for the function $\log(1+x)$ at $x = 0$. Plot the first few of these. What do you notice?

(d) Can you think of an infinitely differentiable function that is not approximated well by its Taylor polynomials, even locally?

Problem 3. Prove the following:

(a) For all $x, y \in \mathbb{R}$:

$$|\sin(x) - \sin(y)| \leq |x - y|.$$

(b) Suppose that $f : \mathbb{R} \to \mathbb{R}$ is differentiable and has a continuous derivative. Prove that the restriction of f to $[0,1]$ is *Lipschitz*, which means that there exists a constant $C \in \mathbb{R}$ such that

$$|f(x) - f(y)| \leq C|x - y|$$

for all $x, y \in [0,1]$.

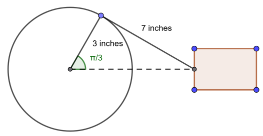

Figure 1: Crankshaft in problem 5

Problem 4. In this problem we will show that an antiderivative of a function on $\mathbb{R}$ is unique up to a constant.

(a) Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function. Prove that f is constant if $f'(x) = 0$ for all $x \in \mathbb{R}$.

(b) Is the same conclusion true for a function $f : (-\infty, 0) \cup (0, \infty)$ be a differentiable function such that $f'(x) = 0$ for all $x \in \mathbb{R}$. Does it follow that f is constant?

(c) Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a function and that there exist functions $F, G : \mathbb{R} \to \mathbb{R}$ for which $F' = f$ and $G' = f$. Prove that there is a constant $C \in \mathbb{R}$ such that $F = G + c$.

2 Challenging problems

Problem 5. In an engine a 7-inch connecting rod attached to a piston is fastened to a crank of radius 3 inches. The crankshaft rotates counterclockwise at a constant rate of 200 revolutions per minute. Compute the velocity of the piston when the crank makes an angle of $\pi/3$ with the x-axis.

Problem 6. In this problem we will show the uniqueness of solutions to ordinary differential equations.

(a) Let $x : \mathbb{R} \to \mathbb{R}$ be a differentiable function and suppose that there exists a constant $C \in \mathbb{R}$ such that
$$x'(t) \leq C \cdot x(t)$$
for all $t \in \mathbb{R}$. Prove that for all $t \geq 0$, we have that
$$x(t) \leq x(0) \cdot e^{Ct}.$$

(b) Suppose that the function $f : \mathbb{R} \to \mathbb{R}$ is *Lipschitz*, which means that there exists a constant $C \in \mathbb{R}$ such that
$$|f(x) - f(y)| \leq C|x - y|$$
for all $x, y \in \mathbb{R}$. Prove that the initial value problem
$$\begin{cases} x'(t) = f(x(t)) \\ x(0) = b \end{cases}$$
can have at most one solution $x : (-a, a) \to \mathbb{R}$ for any $a > 0$ and $b \in \mathbb{R}$.

(c) **Bonus:** Show that the initial value problem in (b) has a solution $x : (-a, a) \to \mathbb{R}$ for *some* $a > 0$. This is harder.